MSDS 6903 Applied Capstone Project D2A

Advanced Model Evaluation and Deployment for Human-Centered AI
in Mental Health Monitoring of Remote Workers

Professor: Syed Raza
Group #3

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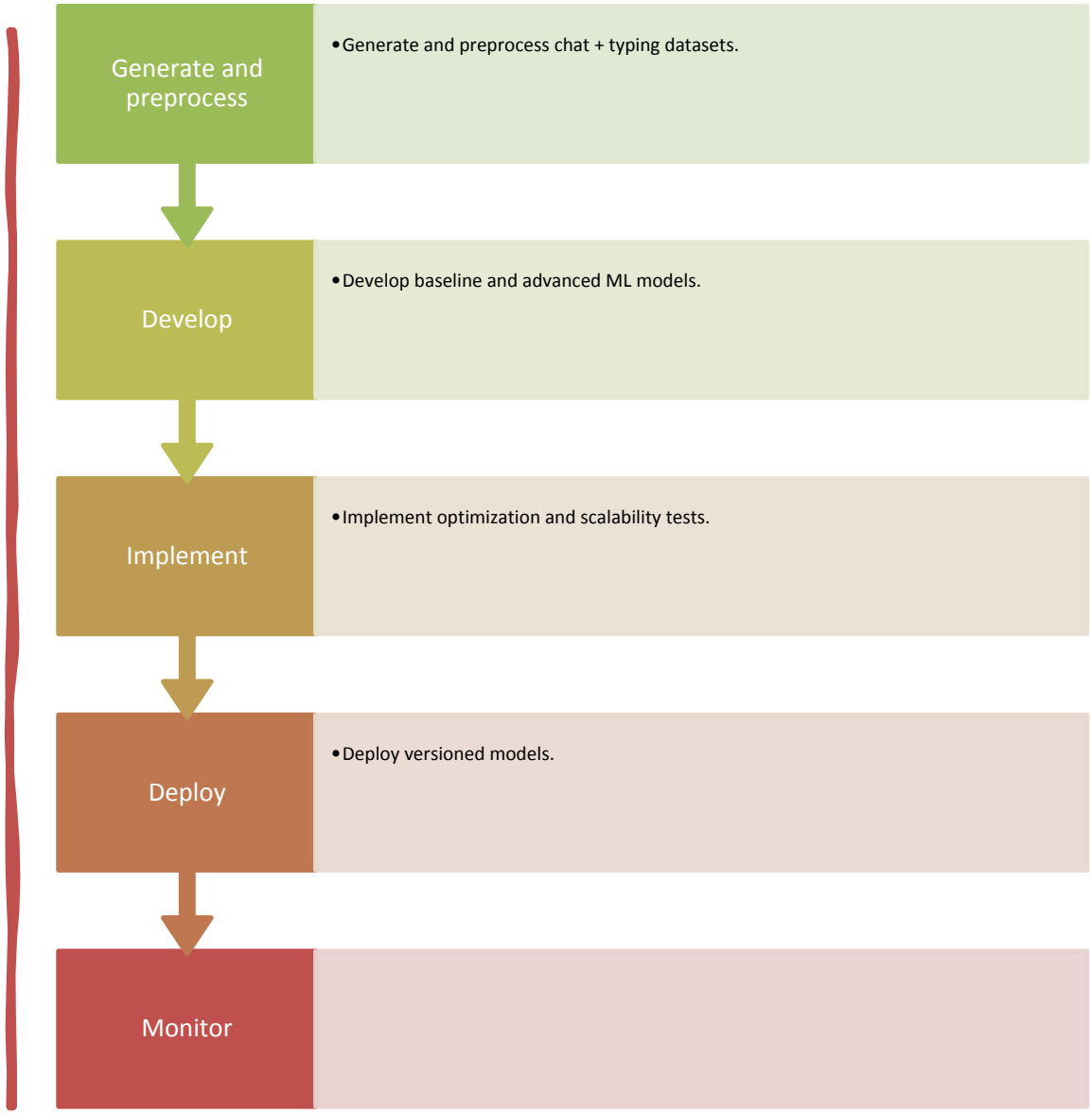
Introduction

- Focus on human-centered AI for mental health monitoring in remote workplaces.
- Built an end-to-end pipeline (data generation → cleaning → modeling → monitoring).
- Goal: Detect engagement risks through chat sentiment + typing behavior.

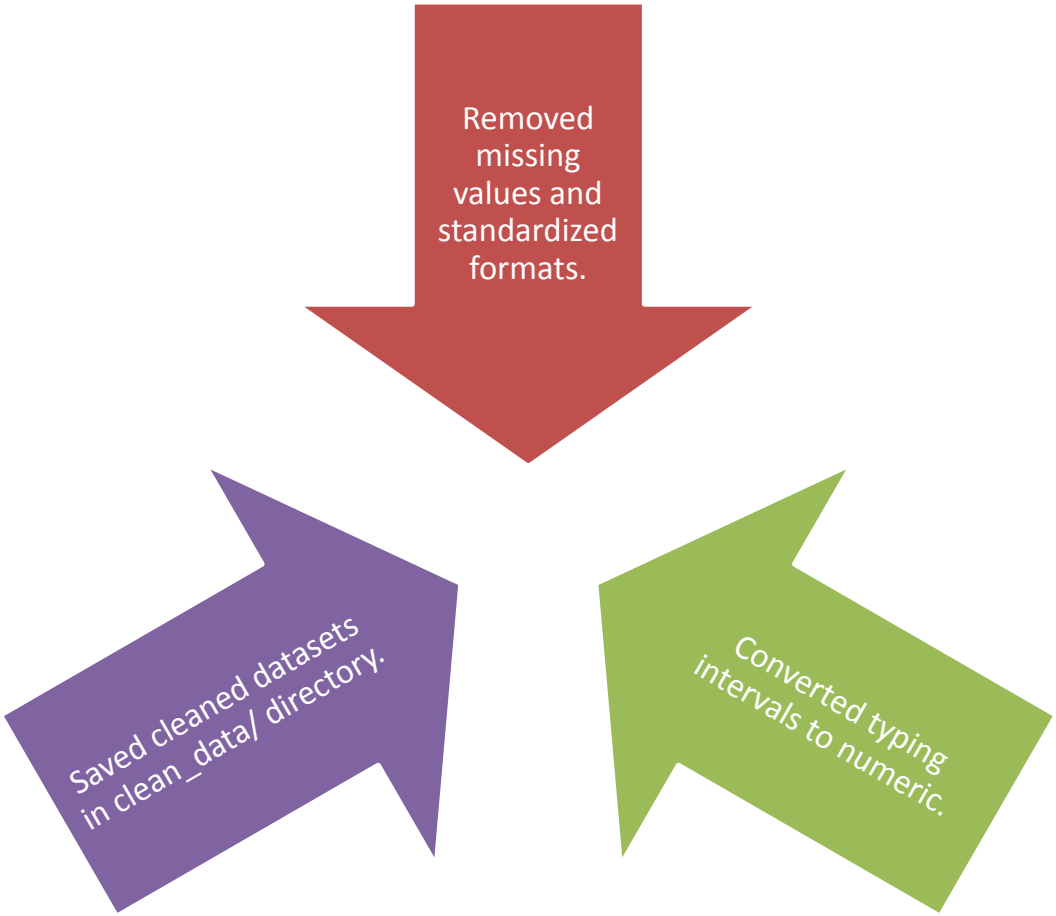
Project Objectives



Data Generation (Assignment 1)

- Chat logs analyzed with VADER sentiment analysis.
- Typing sessions simulated with anomalies.
- Alerts generated for negative patterns or unusual typing behavior.
- Outputs: chat_data.csv, typing_data.csv

Data Cleaning & Storage (Assignment 2)



Baseline Model (Assignment 3)

Features: sentiment score + avg typing ms.

Label: 1 if Negative sentiment or slow typing (>500ms).

Model: Random Forest.

Metrics: Accuracy, Precision, Recall, F1.

Optimization & Scalability (Assignments 4–5)

Optimized Random Forest with 200 estimators.

Benchmarked on 5x replicated data.

Measured training and prediction times.

Saved model for deployment.

Advanced Pipeline (Assignment 6)

TF-IDF VECTORIZER FOR CHAT MESSAGES.

STANDARDSCALER FOR TYPING METRICS.

RANDOM FOREST CLASSIFIER.

VERSIONING: DATASET + MODEL SNAPSHOTS.

Monitoring & Drift Detection

Logged Accuracy, Precision, Recall, F1.

Checked for data drift in class distributions.

Alerts triggered when drift >10%.

Continuous monitoring cycles simulated.

Live Demo (Code Walkthrough)

- Generate raw chat + typing datasets.
- Train baseline and advanced models.
- Evaluate with performance metrics.
- Run monitoring simulation.
- Saved models ready for deployment.

Key Findings

- Engagement risk detectable from multimodal data.
- Random Forest effective for baseline and advanced pipelines.
- Scalability test confirmed model robustness.
- Monitoring pipeline successfully detected drift.

Challenges & Solutions

- Small dataset → used synthetic augmentation.
- Class imbalance → stratified splits.
- Empty batches → added safety checks.
- Text drift detection → proxy with class distributions.
- Versioning → timestamped dataset/model saving.

Conclusion

- Built a complete human-centered AI pipeline.
- Enabled continuous monitoring for reliability.
- Early drift detection improves trustworthiness.
- Future work: deep learning models + real-time alerts.

References

1. Choudhury et al., 2021 - Work-from-anywhere productivity.

2. Moreno & Badia, 2023 - Ethical AI in workplace monitoring.

3. Smith & Lazarus, 2020 - Digital fatigue in hybrid work.

4. Pedregosa et al., 2011 - Scikit-learn: ML in Python.

5. Lu et al., 2018 - Learning under Concept Drift.